BUG LOG

Project Name: Adaptive Multi-Junction Traffic Grid Controller (AMTGC)

During the project, I faced a few issues while running the simulation in ModelSim.

1. **Simulation Timeout:** Initially, the full simulation stopped because of a timeout. I checked the simulation and ran the testbench again. The final run completed successfully.
2. **Waveform Issue:** Initially, the waveform did not display properly. I added the required signals and ran the simulation again. The waveform was displayed correctly.

Final Result: The full regression test passed successfully and displayed `FULL_REGRESSION_PASS`.

ASSUMPTIONS

The following assumptions are considered in this project:

1. The system controls two traffic junctions, Junction A and Junction B.
2. Each junction has North-South and East-West traffic signals.
3. Both junctions follow a predefined traffic signal sequence.
4. Junction B is coordinated with Junction A using the green-wave feature.
5. A single pedestrian arbiter handles requests from both junctions.
6. Emergency input changes both junctions to an all-red condition.
7. Traffic conditions are tested using different input values in the testbench.
8. The project is designed and tested using Verilog HDL and ModelSim.

The project is verified through simulation and is not a real-world traffic control installation.